

CSC 301

Introduction to Java Programming Language

Examination: Semester Examination	Semester: Second Semester
Year: 2024	Duration: 2 hours
Total Questions: 5	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) Explain the concept of Object-Oriented Programming and state its four main principles. (8 marks)
- (b) Differentiate between a class and an object in Java with appropriate examples. (7 marks)

Question 2

- (a) Write a Java program to calculate the factorial of a number using recursion. (10 marks)
- (b) Explain the difference between 'while' and 'do-while' loops in Java. (5 marks)

Question 3

- (a) What is exception handling? Write a Java program that handles division by zero using try-catch blocks. (10 marks)
- (b) List and explain five built-in exception classes in Java. (5 marks)

— End of Examination —